Semester V **B. Tech CSE (AI)**

BCSAI 511: PROLOG PROGRAMMING LAB

0L + 0T + 4P + 2C **MM 100**

List of Experiments:

1. Write a prolog program to calculate the sum of two numbers.

2. Write a prolog program to find the maximum of two numbers.

3. Write a prolog program to calculate the factorial of a given number.

4. Write a prolog program to calculate the nth Fibonacci number.

5. Write a prolog program, insert_nth(item, n, into_list, result) that asserts that result is the list into_list with item inserted as the n'th element into every list at all levels.

6. Write a Prolog program to remove the Nth item from a list.

8. Write a Prolog program to implement append for two lists.

9. Write a Prolog program to implement palindrome(List).

10. Write a Prolog program to implement max(X,Y,Max) so that Max is the greater of two numbers X and Y.

11. Write a Prolog program to implement maxlist(List,Max) so that Max is the greatest number in the list of numbers List.

12. Write a Prolog program to implement sumlist(List,Sum) so that Sum is the sum of a given list of numbers List.

13. Write a Prolog program to implement two predicates evenlength(List) and oddlength(List) so that they are true if their argument is a list of even or odd length respectively.

14. Write a Prolog program to implement reverse(List,ReversedList) that reverses lists.

15. Write a Prolog program to implement maxlist(List,Max) so that Max is the greatest number in the list of numbers List using cut predicate.